Assignment 1: AI System Specification

Course: Introduction to Artificial Intelligence

Application: Fraud Detection System for Digital Payments and Banking Transactions

Task 1: PEAS Specification

Performance Measures:

1) Detect fraudulent transactions correctly.
2) Reduce false fraud alerts for genuine customers.
3) Process transactions quickly.
4) Reduce financial losses due to fraud.

Environment:

The system works in banks, mobile banking apps, online shopping websites, and digital payment platforms. It deals with customers, fraudsters, and different types of online transactions.

Actuators:

1) Block suspicious transactions.
2) Send fraud alerts to customers and banks.
3) Assign a risk score to transactions.
4) Temporarily freeze suspicious accounts.

Sensors

1) Transaction details (amount, time, and merchant).
2) Customer transaction history.
3) Device information (mobile or computer used).
4) Location of the transaction.
5) Login and authentication data.

Task 2: Environment Classification:

1) **Fully Observable vs. Partially Observable**

Choice: Partially Observable

Justification: The system cannot directly know whether a user is a fraudster. It only uses transaction data to make decisions.

2) **Single-agent vs. Multi-agent**

Choice: Multi-agent

Justification: The environment includes customers, fraudsters, and bank security teams.

3) **Deterministic vs. Stochastic**

Choice: Stochastic

Justification: Fraudsters change their methods, so outcomes cannot always be predicted.

4) **Episodic vs. Sequential**

Choice: Sequential

Justification: Previous transactions help the system make future decisions.

5) **Static vs. Dynamic**

Choice: Dynamic

Justification: Transactions happen continuously, and fraud patterns change over time.

6) **Discrete vs. Continuous**

Choice: Continuous

Justification: Transaction amounts and risk scores can have many different values.

Task 3: Critical Analysis

1) **Which dimension is the biggest challenge?**

The Dynamic environment is the biggest challenge because fraudsters continuously develop new ways to commit fraud. The AI system must update itself regularly to detect new fraud patterns.

2) **Utility Function**

Utility Function:

U = 0.7 × Fraud Detection Accuracy − 0.3 × False Positive Rate
This means the system tries to detect more fraud while reducing incorrect alerts for genuine customers.
If the weight of fraud detection accuracy is doubled:
U = 1.4 × Fraud Detection Accuracy − 0.3 × False Positive Rate
the system will focus more on catching fraud. However, it may also block more genuine transactions by mistake.

Task 4: Structured Representation (JSON)

{ "system_name": "Fraud Detection System",
"peas": {
"performance_measures": [
"fraud_detection_accuracy",
"low_false_alerts",
"fast_transaction_processing",
"reduced_financial_loss"
],
"environment": "Banks, mobile banking apps, online shopping sites, and digital payment systems.",
"actuators": [
"block_transactions",
"send_alerts",
"assign_risk_scores",

"freeze_accounts"
],
"sensors": [
"transaction_details",
"customer_history",
"device_information",
"location_data",
"authentication_data"
]
},
"environment_classification": {
"observability": {
"choice": "Partially Observable",
"justification": "The system cannot directly identify fraudsters."
},
"agents": {
"choice": "Multi-agent",
"justification": "Customers, fraudsters, and banks are involved." },
"determinism": { "choice": "Stochastic",
"justification": "Fraud patterns are unpredictable."
},
"episodic_sequential": {
"choice": "Sequential",
"justification": "Past transactions affect future decisions."
},
"static_dynamic":
{ "choice": "Dynamic",
"justification": "Transactions and fraud methods change continuously."
},
"discrete_continuous": { "choice": "Continuous",
"justification": "Transaction values and risk scores vary continuously." }
},
"utility_function": "U = 0.7 * fraud_detection_accuracy - 0.3 * false_positive_rate"
}

Conclusion:

A Fraud Detection System helps banks and payment platforms identify suspicious transactions and reduce financial losses. Since fraud techniques keep changing, the system must continuously learn and improve to provide both security and a good customer experience.

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